
Formalizing AI Scientist Systems: A Unified Framework for Automated Scientific Discovery

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Abstract

The emergence of end-to-end systems capable of automating hypothesis generation, experiment execution, and knowledge synthesis represents a significant change in how science is conducted. Despite rapid empirical progress, AI Scientist systems lack a unifying formal framework, which makes principled comparison across architectures and scientific domains difficult. We propose Servo, a six-component framework (S, G, E, V, M, π) comprising the Search space, hypothesis Generator, Experiment executor, Validator, Memory, and search Policy (π) , grounded in the theory of Partially Observable Markov Decision Processes (POMDPs) and Bayesian Experimental Design (BED). We apply this framework to analyze core AI Scientist systems and their instantiations across seven scientific domains (mathematics, physics, chemistry, materials science, biology, social science, and computer science), and separately examine a complementary artifact infrastructure proposal that addresses the output format and verification layer of such systems. Our analysis surfaces three persistent open problems: (1) the non-automatability of novelty evaluation, which remains dependent on human judgment despite multi-layer automated proxies; (2) the absence of BED-optimal search policies, despite the theoretical availability of tractable approximations; and (3) the underdevelopment of semantic memory M_s , whose absence prevents accumulated experimental experience from improving future search. We show that the completeness of the validator V is the single most consequential design variable: it determines whether closed-loop autonomous operation is feasible in each domain, and current AI Scientist systems can be read as a progression toward more complete validators.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Each system introduces bespoke terminology, making systematic comparison difficult. Three questions that bear directly on deployment remain unanswered: What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?

We address these gaps with three contributions: (1) Servo, a formal framework (S, G, E, V, M, π) connecting AI Scientist systems to POMDP and Bayesian Experimental Design theory (Section 4); (2) analysis of four core systems, an artifact infrastructure

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proposal, and domain-specific instantiations across seven scientific fields (Sections 5, 6); and (3) three open problems that constrain progress across all current systems (Section 8).

Scope. We cover systems whose primary goal is automated scientific discovery through hypothesis-experiment-validation cycles, excluding specialized tools within human-conducted pipelines [Jumper et al., 2021].

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as partially observable Markov decision processes: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O})$, where $\mathcal{A}$ is the action space (hypothesis generation and experiment execution), T is the transition function, R is the reward (scientific value of discovery), and $\mathcal{O}$ is the noisy observation process. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides the normative criterion for the search policy π : select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg \max_{d \in \mathcal{D}} \text{EIG}(d) = \arg \max_d \mathbb{E}[H(\theta) - H(\theta \mid y, d)] \quad (1)$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d . EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022], yet no current AI Scientist system implements it—a gap we identify in Section 8.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop.

Theoretical foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical; we are the first to explicitly apply BED as a design target for AI Scientist search policies, connecting POMDP (structural formulation) to BED (normative policy criterion).

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding as a social model of science; we provide the complementary single-system modular decomposition. Liu et al. [2026] address artifact verification as a V and M_s contribution, discussed in Sections 5.1 and 8.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S, G, E, V, M, \pi) \quad (2)$$

S : Search Space The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \rightarrow h$: Hypothesis Generator A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \rightarrow o$: Experiment Executor A function that executes hypothesis h under protocol P to produce observation o . The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \rightarrow r$: Validator A function that assigns a reward signal r to observation o . We distinguish four layers of V :

$V_{\text{syntax}} : o \rightarrow \{\text{pass, fail}\}$	(deterministic, fast)
$V_{\text{semantic}} : o \rightarrow r \in [0, 1]$	(LLM-based, stochastic)
$V_{\text{empirical}} : o \rightarrow r$	(reproduction-based)
$V_{\text{human}} : o \rightarrow \{\text{novel, significant}\}$	(non-automatizable)

M : Memory Structured as (M_e, M_s, M_p) . M_e (episodic) stores individual records (h_i, o_i, r_i) ; M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π .

$\pi : b(S) \times M \rightarrow h$: Search Policy The decision rule for selecting the next hypothesis. The belief state $b(S)$ is updated after each experiment. In the BED interpretation, $\pi^* = \arg \max_d \text{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned} \pi(b, M) &\rightarrow G(M_s, S) \rightarrow E(h, P) \rightarrow V(o) \\ &\rightarrow M += (h, o, r) \rightarrow b(S) \text{ update} \rightarrow \text{repeat} \end{aligned} \quad (3)$$

We further introduce $H \in [0, 1]$ as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π .

5 Analysis of Core AI Scientist Systems

We apply Servo to four end-to-end systems [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025] and one artifact infrastructure proposal. Table 1 synthesizes the key dimensions; the cross-system pattern is that every architectural advance is a gain in V completeness, and V completeness is what unlocks closed-loop operation.

	Coscientist	AI Sci. (2024)	AI Sci. (2026)	AgentLab
Novelty measure	None	LLM self-score	3-layer	Human-delegated
Loop closed?	No	Yes (greedy)	Yes (tree)	Partial
π type	ReAct	Greedy	Tree search	Human/seq.
V completeness	V_h only	V_s	$V_e + V_s + V_h$	V_s (biased)
M structure	Ephemeral	M_e	$M_e + M_s$	M_e
H (human role)	G	Low	Low	$G + \pi$

Table 1: Comparative Servo analysis of four core AI Scientist systems. V_s =semantic, V_e =empirical, V_h =human.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem: what the output of the AI Scientist loop should look like and how it can be independently verified. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology: `/logic` (claims with falsification criteria), `/src` (executable code), `/trace` (exploration graph: question, decision, experiment, dead_end, pivot), and `/evidence` (raw results). The ARA Seal provides three-level automated verification— V_{syntax} (schema conformance), V_{semantic} (Rigor Auditor, 82.6% detection rate), and $V_{\text{empirical}}$ (sandboxed reproduction)—with V_{human} reserved for significance, novelty, and taste: “judgments of significance, novelty, and taste are reserved

for human reviewers” [Liu et al., 2026]. The motivation is quantitative: only 45.4% of reproduction requirements are fully specified in published papers, and 90.2% of extension costs are consumed by failed exploration [Liu et al., 2026]. The `/trace` layer addresses M_e and M_s : preserved failure traces accelerated agent performance on extension tasks but also anchored more capable agents to prior solution paths—richer M_e does not monotonically improve π .

6 Domain Analysis

Table 2 maps Servo instantiations across seven scientific domains; V completeness covaries with loop closability without exception. To illustrate scale: Aletheia [Feng et al., 2026] attempted 700 Erdős problems, with auditors rating 13/200 graded solutions (6.5%) meaningfully correct; GNoME [Merchant et al., 2023] screened 2.2 million candidate crystal structures via high-throughput DFT. Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024], materials [Merchant et al., 2023], biology [O’Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023], and CS/algorithms [Real et al., 2020, Sartori and Blum, 2024, Jaber and Jaber, 2026].

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Partial	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Cross-domain Servo analysis across seven domains. V reliability refers to agreement with ground truth.

The domain pattern is not coincidental. In mathematics (formal), a noise-free V_{formal} unlocks stable RL: the binary, unambiguous reward cannot be hacked. In chemistry and biology, the loop cannot close because no automated V substitutes for physical measurement (NMR, MS, wet-lab validation). Materials science is the single domain where π approximates BED theory: GNoME’s uncertainty sampling over a tractable GNN posterior is the closest existing deployment to EIG-optimal search. Social science faces a distinct failure mode: when G and E share the same underlying model, the evaluator systematically confirms the generator’s implicit biases [Aher et al., 2023].

7 Illustrative Simulation: Validator Coverage and False Novelty

To make the consequences of V incompleteness concrete, we run a minimal agent-based simulation (20 agents, 200 hypotheses, 40 latent classes, 30 seeds) in which agents estimate novelty via a coverage membership test and share memory only through their communication topology—the primary independent variable. Table 3 formalizes the four conditions as levels of V completeness.

Results. Figure 1 shows FNR (fraction of non-novel hypotheses marked novel) across conditions. Because memory sharing is set-union, broader connectivity structurally reduces FNR—an analytic consequence, not an emergent finding. FNR decreases from 0.955 ± 0.028

Table 3: Simulation conditions. Each operationalizes a level of V completeness via mean degree k .

Condition	V -completeness	k	Graph	Coverage
Isolated	Incomplete	0	No edges	Own memory only
Lattice	Partial	2	Ring	2-hop neighborhood
Medium	Improved	4	Watts–Strogatz	Local + cross-links
Shared	Complete	$n-1$	Complete	Full community memory

(isolated) to 0.008 ± 0.012 (shared); varying classifier accuracy 0.60–0.90 moves FNR by ≤ 0.035 , a $27\times$ smaller range than the cross-topology span. A non-trivial consequence emerges at $\geq 10\%$ prior seeding: isolated agents achieve higher balanced accuracy than lattice (0.564 vs. 0.529), as connectivity accelerates memory saturation. The analogy to epistemic-network models [Zollman, 2007, 2010] is phenomenological only.

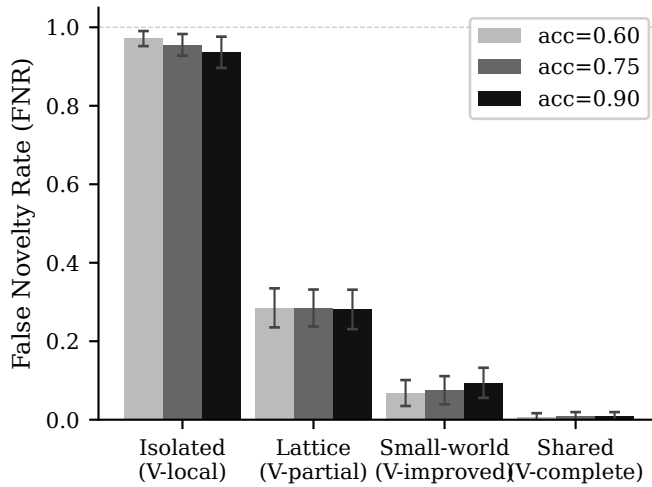


Figure 1: FNR by V -completeness level and classifier accuracy (± 1 std, 30 seeds). Cross-topology span: 0.947; within-condition accuracy range: ≤ 0.035 .

Limitations. Synthetic agents with a ground-truth oracle; agents do not generate hypotheses or update beliefs. The simulation is an existence illustration of V incompleteness—the FNR-topology ordering is structurally determined and cannot produce any other outcome.

8 Open Problems

Our analysis identifies three open problems that persist across all systems and domains.

8.1 The Non-Automatability of Novelty

Novelty cannot be reliably evaluated by automated means: every system from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation) converges on this conclusion. Two failure modes are documented: LLM self-evaluation bias (systematic over-estimation; hallucinated results) and pretraining contamination (systems may “discover” results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal [Liu et al., 2026] demarcates the automated frontier—82.6% detection of structural violations with a 22% blind spot for orphaned experiments—but cannot cross the novelty boundary. Concretely: Agent Laboratory’s LLM reviewer over-estimated quality by +2.3 points on average; The AI Scientist (Sakana) hallucinated ablation tables; and Aletheia audited 700 Erdős problems and found only 6.5% of graded solutions meaningfully correct despite surfacing many superficially plausible candidates.

Falsification criterion: Refuted if an automated V achieves human-level novelty discrimination (inter-rater agreement ≥ 0.80) on a contamination-controlled benchmark spanning ≥ 3 distinct scientific domains.

8.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no current system implements it: all deployed policies are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty sampling approximates EIG over a tractable GNN posterior—but only because DFT provides a well-calibrated V and the hypothesis space admits a learnable probabilistic model, conditions absent from LLM-based systems where the “posterior” is implicit in model weights [Foster et al., 2021]. Concretely, every system in Table 1 degrades to heuristic π : Coscientist selects the next tool reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed sequential schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid.

Falsification criterion: Refuted if any AI Scientist system implements a policy that explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ in a domain where the posterior is not tractable from first principles—ruling out tractable-model proxies such as GNoME’s deep-ensemble policy.

8.3 The Underdevelopment of Semantic Memory M_s

Current systems store raw experiment records (M_e) but do not distill them into transferable patterns (M_s): a system with 1,000 experiments does not search more intelligently than one with 10. The ARA /*trace* layer [Liu et al., 2026] is the closest approximation, but traces accelerated extension performance while also anchoring capable agents to prior solution paths— M_s structure must be paired with selective forgetting mechanisms to be beneficial. Concretely, GNoME’s GNN—one of the few systems where $G(M_s, S)$ genuinely improves—accumulates domain-specific thermodynamic knowledge after 460K DFT calculations, yet this knowledge does not transfer to chemistry or biology; the system is no more capable in any domain it has not explicitly trained on.

Falsification criterion: Refuted if accumulated cross-experiment knowledge demonstrably improves G -generated hypothesis quality across task instances in any single domain, measured by a pre-specified metric.

The three open problems are not independent. A system with complete V and tractable M_s would, by construction, be able to implement an EIG-optimal π : complete V provides a calibrated, hack-resistant reward signal, and developed M_s maintains the hypothesis-space model that EIG requires. They therefore converge on a single structural gap, and progress on any one unlocks disproportionate gains toward the other two. They are also precisely the necessary conditions for a collective AI co-scientist [Taniguchi et al., 2025]: $V_{\text{collective}}$ requires solved novelty automatability; a principled collective π requires closed BED–practice gap; and the shared semantic memory w requires developed M_s . ARA [Liu et al., 2026] may be the most tractable entry point, simultaneously externalizing M_s and calibrating the sub-novelty layers of V .

9 Discussion

Our analysis establishes a design hierarchy: V choice precedes all other decisions, as it determines closed-loop feasibility and what form π can take. The collective extension [Taniguchi et al., 2025] requires that the three open problems be solved first: $V_{\text{collective}}$ requires novelty automatability; a principled collective π requires closing the BED–practice gap; developed M_s enables the shared semantic memory w . ARA [Liu et al., 2026] may be the most tractable entry point as a structured artifact protocol for w . Limitations: Servo abstracts domain-specific cost structures of E , non-stationarity of S , and the social layer that Taniguchi et al. [2025] formalize.

10 Conclusion

We introduced Servo (S, G, E, V, M, π) , a formal framework connecting AI Scientist systems to POMDP and BED theory, and applied it to four core systems, one artifact infrastructure proposal, and seven scientific domains. The central finding is that V completeness is the primary determinant of closed-loop feasibility: every architectural advance across surveyed systems is ultimately a gain in V completeness. The three open problems—novelty non-automatability, the BED–practice gap, and M_s underdevelopment—converge on a single organizing question: how do we build validators and memory structures that are simultaneously complete enough to support closed-loop operation and structured enough to support principled search?

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a Research Stage Assessment (items 1–4) rating AI involvement and iteration effort, and an AI System Documentation section (items 5–6).

Research Stage Assessment

1. Hypothesis development.

Answer: [B]

Iteration Effort: [Medium]

Explanation: The core Servo framework tuple (S, G, E, V, M, π) , the V -completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI’s elaborations.

2. Experimental design and implementation.

Answer: [C]

Iteration Effort: [High]

Explanation: The agent-based simulation (20 reviewer agents, four topology conditions, FNR metric) was designed and implemented in Python by a frontier LLM agent under human direction. Multiple rounds of adversarial review and code revision were required; the human author specified the research question and directed structural decisions.

3. Analysis of data and interpretation of results.

Answer: [C]

Iteration Effort: [Medium]

Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V -completeness thesis emerged from the human author’s synthesis.

4. Writing.

Answer: [C]

Iteration Effort: [High]

Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §8), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. AI systems used.

Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. Observed AI Limitations.

Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance of simulation results that were analytically entailed; adversarial review rounds were needed to correct framing. (3) Context drift:

over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems) and acknowledge the simulation as illustrative rather than empirically novel. The V-completeness thesis is presented as a cross-domain pattern, not a proven theorem.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are discussed in §9 (framework abstracts domain cost structures, social layer, non-stationarity) and in §7 (simulation is an existence illustration, not empirical measurement).

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [NA]

Justification: The paper makes no formal theorem claims. The V-completeness thesis is a cross-domain empirical pattern supported by natural comparison, not a proven theorem.

4. Experimental result reproducibility

Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: Simulation parameters (20 agents, 200 hypotheses, 40 latent classes, 30 seeds, 4 topology conditions) and the FNR metric are fully specified in §7. Simulation code will be released upon acceptance.

5. Open access to data and code

Question: Does the paper provide open access to data and code?

Answer: [No]

Justification: Simulation code is not yet publicly released but will be made available upon acceptance. No external datasets were used; all analyzed papers are publicly accessible.

6. Experimental setting/details

Question: Does the paper specify all details necessary to understand the results?

Answer: [Yes]

Justification: All simulation hyperparameters are stated in §7. The cross-domain analysis methodology (component extraction from primary sources) is described in §5 and §6.

7. Experiment statistical significance

Question: Does the paper report error bars or statistical significance information?

Answer: [Yes]

Justification: Figure 1 reports ± 1 std across 30 seeds for all FNR values. The paper explicitly notes that the within-condition accuracy range (≤ 0.035) is $27\times$ smaller than the cross-topology span (0.947), presented descriptively.

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [NA]

Justification: The simulation runs in seconds on standard laptop hardware (pure Python, no GPU required). Compute resources are not a meaningful constraint for this experiment.

9. Research Ethics

Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc’s AI authorship policies apply?

Answer: [Yes]

Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. Broader impacts

Question: Does the paper discuss both potential positive and negative societal impacts?

Answer: [Yes]

Justification: §9 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the M_s underdevelopment problem and novelty non-automatability).